

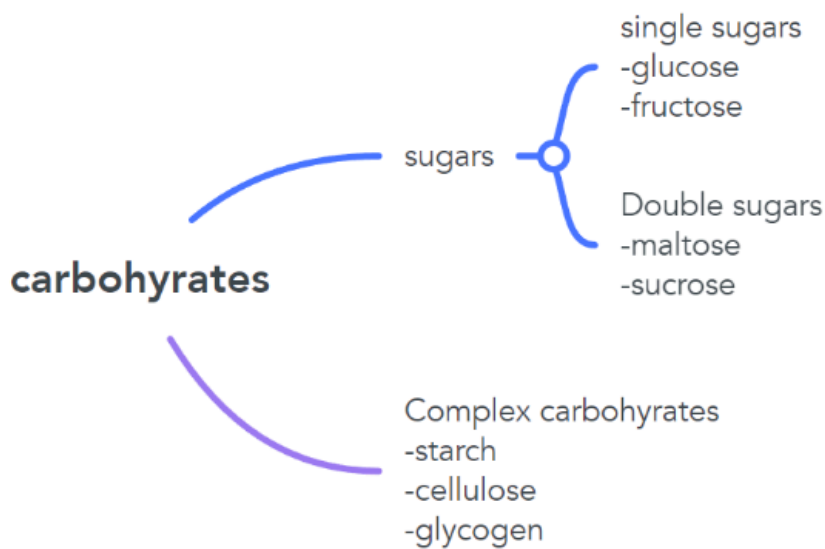
Carbohydrates

Definition: Carbohydrates are organic molecules made of the elements carbon, hydrogen, oxygen. hydrogen and oxygen atoms are present in the ration 2:1

#BiologyDefinitions

- Glucose: $C_6H_{12}O_6$
- Sucrose (Cane sugar): $C_{12}H_{22}O_{22}$

Classification of carbohydrates:



Sugars:

- sweet and soluble in water
- when dissolved in water $\Rightarrow$ it lowers WP of solution
- sugars provide us with the energy to do work and carry out activities

Single Sugars (Monosaccharides)

$\rightarrow$ basic unit of a carbohydrate

$\rightarrow$ can pass through cell membrane and be absorbed into cells

common single sugars: glucose and fructose

Both have chemical formula of $C_6H_{12}O_6$

$\Rightarrow$ atoms are arranged differently within molecules $\Rightarrow$ diff. arrangement gives rise to different chemical and physical properties

Glucose $\Rightarrow$ found in both plants and animals

Fructose $\Rightarrow$ common plants, rare in animals

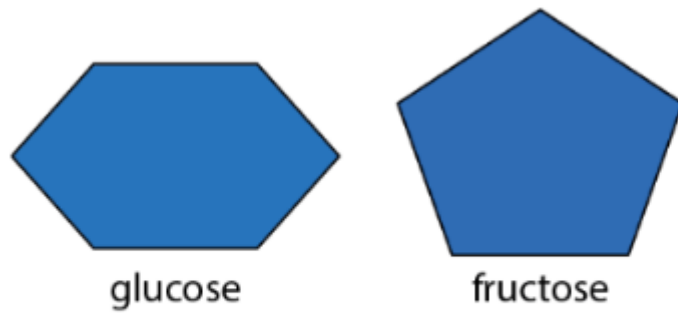


Figure 3.4 Structures of glucose and fructose

Double Sugars (Disaccharides)

⇒ formed when 2 single sugars join together

common double sugars ⇒ maltose and sucrose

maltose (malt sugar) ⇒ occurs in germinating grains

formed when 2 glucose molecules bond together



Sucrose molecule ⇒ made up of 1 fructose and 1 glucose molecule joined together

Glucose + fructose → sucrose

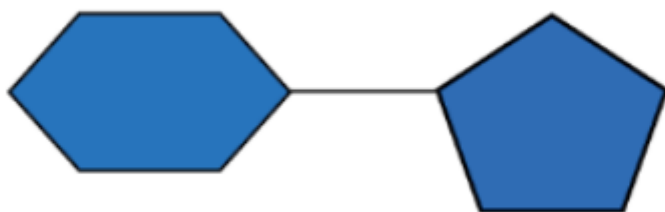


Figure 3.5 Structure of sucrose

Double sugar ⇒ can be split into 2 single sugars molecules ⇒ using an organic molecule called enzymes.



How can we test for sugars?

⇒ we can use benedicts test to test for sugars.

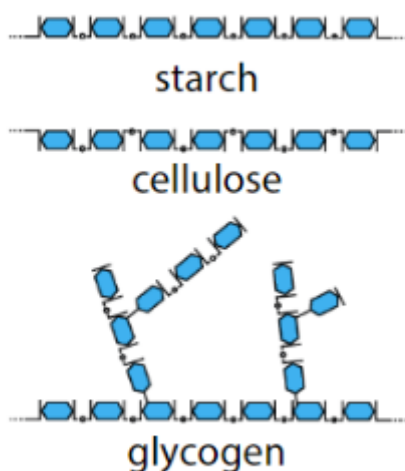
refer to → **Food Tests**

Complex Carbohydrates (Polysaccharides)

→ made up of many similar molecules of simple sugars joined together to form a large molecule

Starch, glycogen, cellulose ⇒ complex carbohydrates made up of numerous glucose molecules.

- glucose molecules linked in different ways ⇒ giving rise to diff structures
- thus they also have diff chemical and biological properties
- also have diff roles



Complex Carbohydrate	Structure	Role	Occurance
Starch	made up of several thousand glucose molecules joined	→ storage form of carbohydrates in plants → when required it	→ storage organs of plants ⇒

Complex Carbohydrate	Structure	Role	Occurance
	together in long straight chains	can be digested to form glucose to provide energy for cell activities.	potato, tubers, tapioca
Cellulose	cellulose molecules are made of many glucose molecules joined together ⇒ made up of straight chains of glucose. ⇒ bonds between glucose units are different from that in starch	→ cellulose cell wall prevents cells from bursting or damage. → cellulose cannot be digested in our intestines → serves as dietary fibre that prevents constipation.	Present in cell wall of plant cells
Glycogen	glycogen is a branched molecule. → chains of glucose that are highly branches	Storage form of carbohydrates in mammals. ⇒ when needed it can be digested to glucose to provide energy for cell activities.	muscles and liver of mammals

Glycogen and starch are stores of glucose

plants → store glucose in the form of starch

animals → cannot make or store starch in their bodies (however, they get most carbohydrates from plants in form of starch)

animals main store of energy → glycogen.

- stored mainly in liver and muscles



glucose and starch are suitable storage materials in cells as:

- they are insoluble in water ⇒ do not reduce WP of cells

- large molecules $\Rightarrow$ cannot diffuse through cell membranes of cells $\Rightarrow$ will not be lost from cell
 - Easily broken down into glucose when needed $\Rightarrow$ eg: tissue respiration
 - molecules have compact shapes $\Rightarrow$ occupy less space than all the glucose molecules that make up glycogen or starch molecules.
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Function of Carbohydrates

Functions of Biological Molecules

Quick Reference:

- substrate for respiration to provide energy for cell activities(immediate source of energy)
 - Formation of nucleic acids - DNA
 - form supporting structures - cell walls in plants
 - converted to other organic compounds like proteins and fats
 - synthesize lubricants $\Rightarrow$ mucus $\Rightarrow$ which is made up of a carbohydrate and a protein
 - synthesize nectar in some flowers
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Test For Starch

Refer to: [Food Tests](#)
